Digital Clock

By Animesh Bodhak

This project is a digital clock that is based purely on synchronous logic circuitry. A top-level view of the design is as follows:

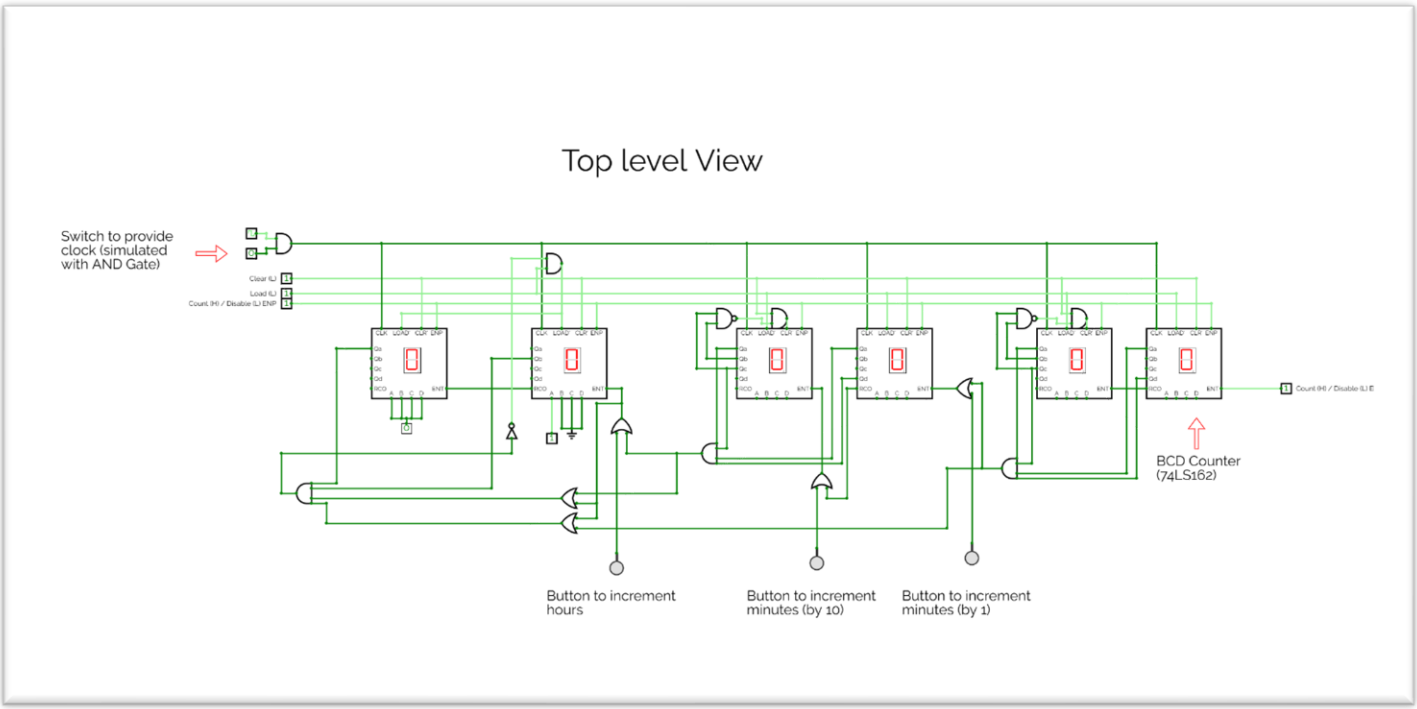


Figure 1: Top-level view of circuit from Simulation (on CircuitVerse)

Links

The following are useful links regarding this project:

- GitHub Repository Link: https://github.com/codevortex/Digital-Clock
- CircuitVerse Simulation Link: https://circuitverse.org/users/434705/projects/digital-clock-using-ti-chips
- Simulation YouTube Video: https://youtu.be/qJywzU_hOa8

Clock Generation

First and foremost, the clock is generated using a Duty Cycle Square Wave Oscillator circuit (provided by Scott Wang):

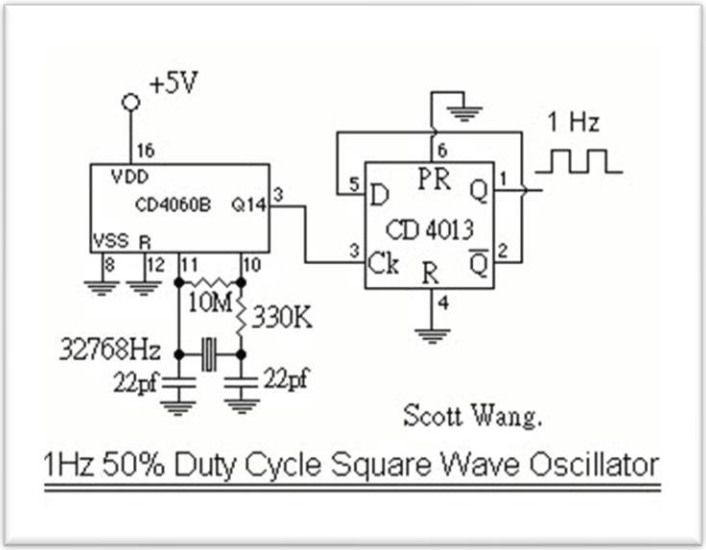


Figure 2: 1Hz Duty Cycle Square Wave Oscillator Circuit by Scott Wang

This circuit uses a 32.768kHz crystal in combination with two 22pf Capacitors, one 10M Ω Resistor, and one 330K Ω resistor to generate a 32.768kHz pulse. This pulse is first brought down to 2Hz by the CD 4060B 14-stage through hole binary counter. The CD 4013B CMOS Dual D-Type Flip-Flop further decreases the frequency to 1Hz.

ENP, ENT, $\overline{\text{Clear}}$, $\overline{\text{Load}}$ Inputs

The 1Hz clock generated by the aforementioned crystal circuitry is fed to six 74LS162 BCD Counter ICs. A high level at the ENP, ENT, $\overline{\text{Clear}}$, and $\overline{\text{Load}}$ inputs for the chip allows the BCD counter to increment on the next clock pulse (positive rising edge). When the counter resets from 9 to 0, a pulse is sent through the RCO output. This pulse can be used to increment the next BCD counter.

Because the ENP input is used for carry look-ahead circuitry (which is not needed for the digital clock), it is set to a constant high level.

The ENT input is used to increment the BCD counter. By making the ENT input a high or low level, the timing of increment is controlled. For example, to increment the units digit of hours, the ENT input is high only when the seconds is 59 (i.e., 1010 = 5, 1001 = 9). Buttons are used in conjunction with the OR gate to allow for a manual increment in the BCD counter.

A low-level at the $\overline{\text{Clear}}$ input causes the BCD counter to clear on the next clock pulse irrespective of the ENP or ENT inputs. To make this clear function synchronous throughout the digital clock, the $\overline{\text{Clear}}$ input is connected to each BCD counter. The $\overline{\text{Clear}}$ input is also used to reset the BCD counter to 0 after it passes 5 (in minutes and hours).

A low-level at the $\overline{\text{Load}}$ sets the clock to 01:00:00 on the next clock pulse regardless of ENP or ENT inputs. $\overline{\text{Load}}$ is also used to reset the hour(s) BCD chip from 12 to 01.

Physical Board

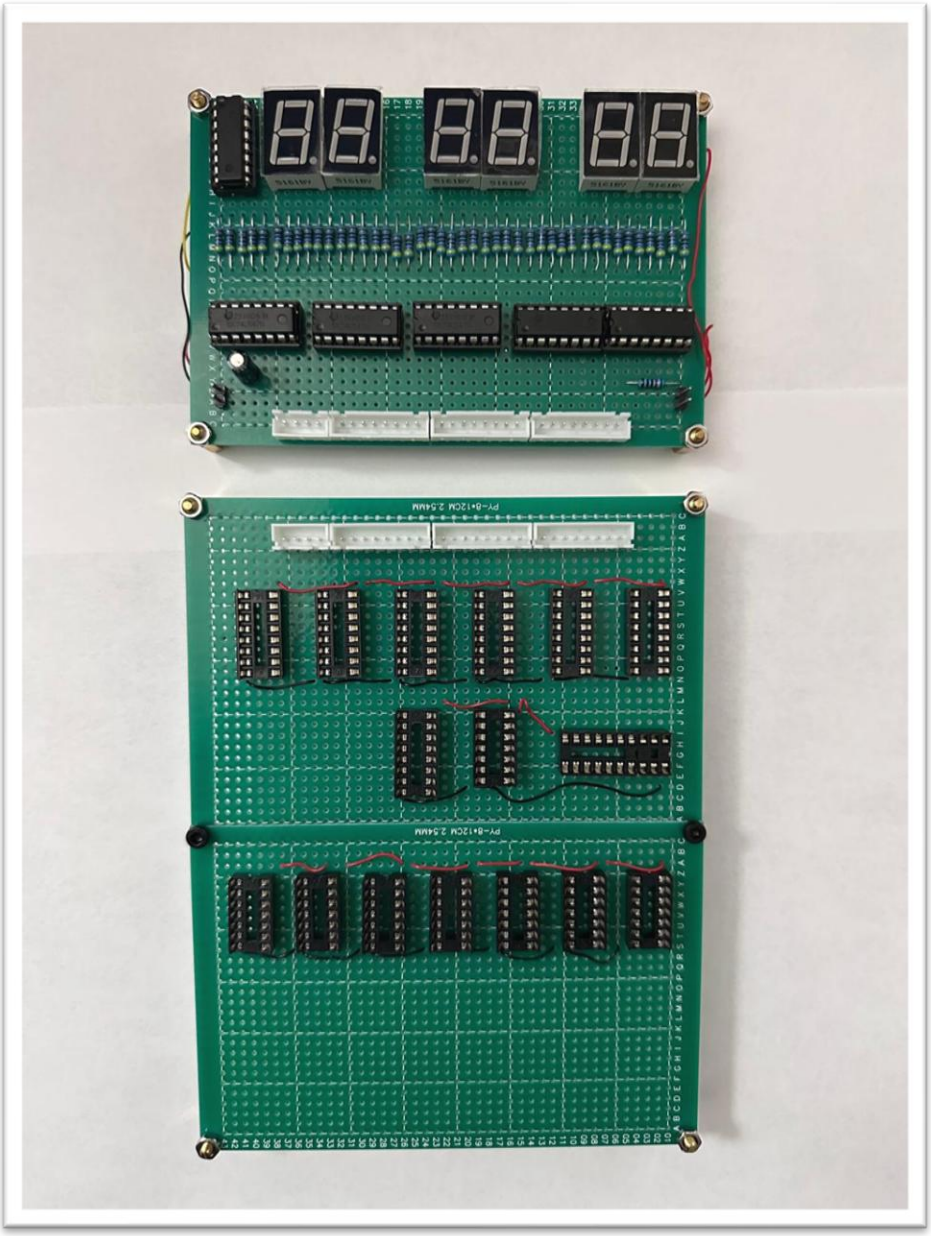


Figure 3: Layout of Digital Clock Board(s)

The current board is currently being soldered. The layout of the board is as follows:

The board is divided into 3 main parts: a display portion (top), a clock generation portion (middle), and a logic portion (bottom). As of September 9, 2026, the display portion of the board is complete; the other two portions are being soldered.

The display portion (top) uses six 7-segment LEDs (common anode) to show the time in HH: MM: SS. Six 74LS47 BCD to 7-Segment Decoder/Driver ICs are used to drive these LEDs.

The clock generation portion (middle) uses the aforementioned clock circuitry (by Scott Wang) to generate a 1Hz clock. The CD4013 CMOS IC, however, cannot drive the six 74LS162 BCD Decade Counters. A buffer is thus used via the 74LS244 Octal Buffers and Line Drivers w/ 3-state outputs IC (bottom right of middle board).

The logic portion (bottom) implements the logic circuitry seen in the simulation using the 74LS00 Quad 2-Input NAND, 74LS08 Quad 2-Input AND, 74LS21 Dual 4-Input AND, and 74LS32 Quad 2-Input OR Gates. Not shown above are the incrementing buttons and $\overline{\text{Load}}$ and $\overline{\text{Clear}}$ switches, along with the debouncing circuits that accompany them.